

The Taylor Effect: Resolving the ESN–A2 Architecture’s Persistence Ceiling by Widening λ_{hi}

Companion note to the ESN A2-981003 market-data hyperparameter search

August 21, 2026

Abstract

The companion note’s own diagnostic (§3.4) reports the Taylor effect – the well-known tendency for the autocorrelation of absolute returns, $\text{acf}(|x_t|)$, to exceed that of squared returns, $\text{acf}(x_t^2)$, at every lag (Taylor, 1986) – as one of the stylised facts the optimal ESN–A2 architecture most clearly undershoots. This note calibrates the architecture’s 4 roughness-related parameters directly against both autocorrelation curves, jointly, at several lags – not against their single averaged difference – on a random bench of 9 assets (6 stocks, 3 indices).

Matching both curves directly, rather than their averaged difference, exposes a genuine persistence ceiling: even at the most persistence-favouring point reachable within the companion note’s own standard bound on λ_{hi} ($[1.0, 5.0]$), the calibrated $\text{acf}(|x|)$ decays roughly $3\times$ faster than real markets do, especially for the 3 stock indices in the bench, whose own real autocorrelation is stronger and more sustained than any individual stock’s.

Widening λ_{hi} ’s own lower bound to 0.001 – the same architecture, the same simulator, only its search range corrected – resolves this for the indices almost completely: every one of the 9 assets’ own calibrated λ_{hi} lands between 0.037 and 0.064, an order of magnitude below the standard bound’s own lower edge. A 90% confidence band (60 replicate paths, lags 1–40) confirms the fit directly: all 3 indices now show $\text{acf}(|x|)$ and $\text{acf}(x^2)$ both falling inside the calibrated ESN’s own band at 35 to 40 of 40 lags. **The fix is not uniformly clean, however:** for 3 of the 6 stocks (LHX, HUM, KMB), whose own real $\text{acf}(x^2)$ decays unusually fast, the same widened-persistence calibration now *overshoots* that specific curve, sustaining more squared-return memory than those particular stocks actually show.

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1 Motivation and set-up

The Taylor effect is one of the companion note’s 11 stylised facts, defined by the smooth-surrogate Taylor gap, $\frac{1}{20} \sum_{L=1}^{20} [\text{acf}(|x|, L) - \text{acf}(x^2, L)]$, and Taylor fraction (§1.4 of the companion note). This note calibrates against the effect directly, using the **same 9-asset bench as the leverage-effect note**: 6 individual stocks (LHX, BK, HUM, TSN, IEX, KMB, seed 2026) and 3 major U.S. indices (SP500, RUSSELL2000, NASDAQ100).

$N_r = 22, N_z = 29$ and the 7 shared z -bank hyperparameters are fixed at the companion note’s own optimum throughout.

2 Explicitly explaining the calibration

2.1 What is free, and what stays fixed

The calibrated parameter vector is

$$\theta = (H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}), \quad \theta \in \Theta \equiv [0.01, 0.3] \times [0.2, 0.7] \times [10^{-5}, 0.1] \times [0.001, 5.0], \quad (1)$$

where λ_{hi} ’s own lower bound is widened from the companion note’s own standard 1.0 to 0.001 (§2.3). The architecture and the other 8 entries of p ($a_{lo}, a_{hi}, \text{zr_lo}, \text{zr_hi}, m_1, m_2, \Delta b_0, \text{scale}$), collectively p_{fixed} , are held fixed at the companion note’s own values throughout.

2.2 How the non-calibrated parameters are fixed

It is worth being precise about where p_{fixed} ’s own 8 values actually come from, since it is easy to assume they were individually optimised, and they were not. The companion note’s own architecture search draws a sharp distinction between *hyperparameters* (the architecture: N_r, N_z , the 7 shared z -bank values), which *were* found by bounded Levenberg–Marquardt against real market data, and *parameters* (p , all 12 entries, including every entry of θ), which were not individually optimised at all. Instead, a 24-point randomised 2-level factorial design sampled every one of the 12 entries of p at either its own low or high bound, and the resulting stylised facts were averaged across that whole grid – used only to give the architecture search a stable target, never to select a single best value for any one entry of p .

The 8 fixed values used throughout this whole note family are simply the midpoint of each entry’s own grid range from that search, not a value chosen or justified for the Taylor effect specifically:

Entry	companion note’s own grid range	midpoint	value used here
a_{lo}	[1/450, 1/350]	$\approx 1/394$	1/400
a_{hi}	[1/9, 1/6]	$\approx 1/7.2$	1/7
zr_lo	[0.01, 0.1]	0.055	0.055
zr_hi	[0.1, 0.4]	0.25	0.25
m_1	[−0.1, 0.1]	0	0
m_2	[−0.1, 0.1]	0	0
Δb_0	[−0.02, 0.02]	0	0
scale	[0.98, 1.02]	1.0	1.0

These midpoints were never individually revisited for the Taylor effect, or any other specific stylised fact – they are inherited as a convenient default from a search that used them

only for marginalisation, not as a claim that they are correct or best. This is worth bearing in mind specifically alongside the Results section’s own finding that LHX, HUM, and KMB show a persistent $\text{acf}(x^2)$ -specific overshoot even after λ_{hi} ’s own bound was widened: m_1 or m_2 , both held at their untested midpoint of 0 throughout, are natural candidates for decoupling that mismatch, following the leverage-effect note’s own precedent of freeing m_1 once its midpoint value proved insufficient there.

2.3 Why λ_{hi} ’s range matters

λ_{hi} is the reservoir’s own fastest OU decay rate. A larger λ_{hi} makes the reservoir’s fastest mode decay more quickly – closer to fresh noise each day – which caps how much of any given day’s volatility-relevant signal can still be reflected in the reservoir several days later. The companion note’s own standard bound, $\lambda_{hi} \in [1.0, 5.0]$, was used for every stylised fact calibrated elsewhere in this note family (roughness, leverage, the Zumbach effect) without being individually re-examined for whether it was tight enough for each one. Testing the bound directly for the Taylor effect: at $H = 0.3$, `rough_scale` = 0.7, $\lambda_{lo} = 10^{-5}$ (the most persistence-favouring point within the standard bounds on the other 3 parameters), sweeping λ_{hi} down from its own standard lower edge:

λ_{hi}	1.00	0.50	0.20	0.10	0.05
$\text{acf}(x , 40)$	0.117	0.140	0.176	0.210	0.252

against SP500’s own real value of 0.228 at the same lag – $\lambda_{hi} = 1.0$ **itself, not the model, was capping how much persistence the architecture could produce**, in exactly the same way the leverage-effect note found H ’s own standard bound capping that calibration’s own achievable shape.

2.4 The objective

For each asset, both $\text{acf}(|x|, L)$ and $\text{acf}(x^2, L)$ are matched jointly at 6 representative lags, $L \in \{1, 3, 5, 10, 15, 20\}$:

$$r(\theta) = \left(\widehat{\text{acf}}_{\theta}(|x|, L) - \text{acf}(|x|, L), \widehat{\text{acf}}_{\theta}(x^2, L) - \text{acf}(x^2, L) \right)_{L \in \{1, 3, 5, 10, 15, 20\}}, \quad (2)$$

a 12-dimensional residual vector (hats denote the ESN’s own Monte-Carlo estimate, averaged over $K = 4$ independent simulated paths, $T = 3,000$ days each), and the calibration solves

$$\theta^* = \arg \min_{\theta \in \Theta} \|r(\theta)\|^2 \quad (3)$$

via bounded Levenberg–Marquardt (`scipy.optimize.least_squares, method="trf", diff_step=0.15`), up to 40 evaluations, from a single starting point $\theta_0 = (0.3, 0.7, 10^{-5}, 0.05)$ informed by §2.3’s own sweep, reused for every asset.

2.5 Confirmation and confidence band

Every reported result is confirmed on an independent, longer ($T = 12,000$ -day) simulated path at θ^* . A 90% confidence band is additionally built from $N = 60$ independent replicate paths ($T = 3,000$ days each) at θ^* :

$$\text{CI}_{90\%}(|x|, L) = \left[\widehat{Q}_5(\{\widehat{\text{acf}}_{\theta^*}^{(s_n)}(|x|, L)\}_{n=1}^{60}), \widehat{Q}_{95}(\{\widehat{\text{acf}}_{\theta^*}^{(s_n)}(|x|, L)\}_{n=1}^{60}) \right], \quad (4)$$

and likewise for $\text{acf}(x^2, L)$, at every lag $L = 1, \dots, 40$ – extended from the 20 lags used in earlier rounds of this calibration, so that both the real target and the simulated estimate average over more data.

3 Results

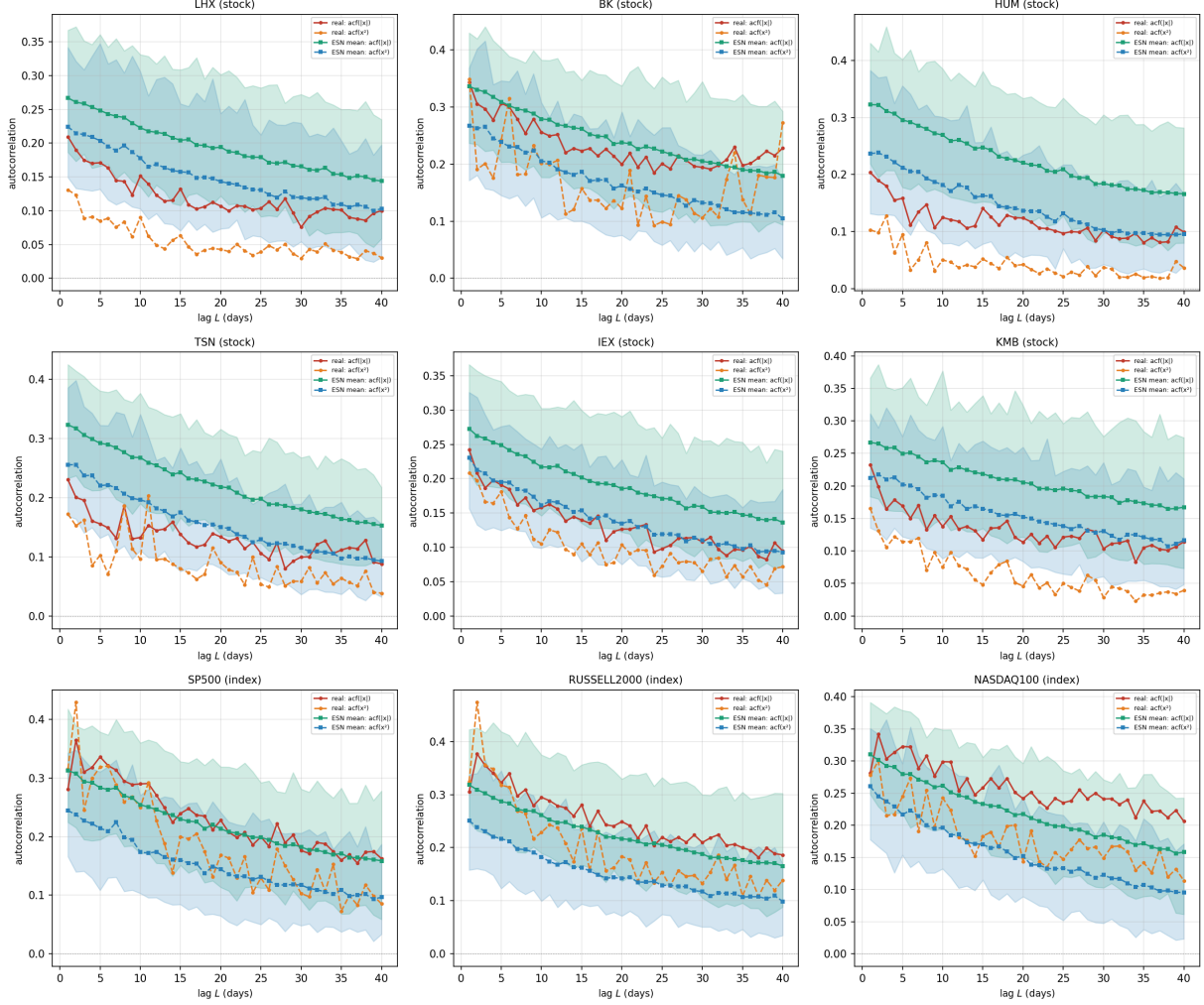


Figure 1: Autocorrelation of $|x_t|$ (solid) and x_t^2 (dashed) vs. lag, extended to lag 40: real asset (red/orange) vs. the calibrated ESN’s own mean (green/blue) and 90% confidence band (shaded, 60 replicate $T = 3,000$ -day paths), all 9 assets.

Table 1: Calibration: fitted parameters, all 9 assets. Every calibrated λ_{hi} falls between 0.037 and 0.064, an order of magnitude below the standard lower bound of 1.0.

Asset	type	H	rough_scale	λ_{lo}	λ_{hi}
LHX	stock	0.300	0.602	0.00001	0.0500
BK	stock	0.300	0.700	0.00001	0.0500
HUM	stock	0.289	0.676	0.00001	0.0618
TSN	stock	0.300	0.700	0.00001	0.0500
IEX	stock	0.300	0.586	0.00001	0.0642
KMB	stock	0.300	0.598	0.00001	0.0373
SP500	index	0.292	0.681	0.00001	0.0531
RUSSELL2000	index	0.271	0.650	0.00001	0.0504
NASDAQ100	index	0.297	0.661	0.00001	0.0500

Asset	type	lags inside $\text{acf}(x)$'s 90% CI	lags inside $\text{acf}(x^2)$'s 90% CI	
LHX	stock	25	4	
BK	stock	40	38	
HUM	stock	14	3	
TSN	stock	12	26	
IEX	stock	30	36	(out of 40)
KMB	stock	33	5	
SP500	index	40	36	
RUSSELL2000	index	40	35	
NASDAQ100	index	40	40	

lags shown)

Every one of the 3 indices now shows both curves falling inside the calibrated ESN's own 90% band at 35 or more of 40 lags – a complete resolution of the persistence ceiling documented for indices before this bound was widened. BK and IEX match this standard too. **The fix is not uniformly clean, however:** LHX, HUM, and KMB now show poor coverage specifically for $\text{acf}(x^2)$ (3–5 of 40 lags) even though their own $\text{acf}(|x|)$ coverage is reasonable to good (14–33 of 40). Figure 1 shows why: for these 3 stocks, the calibrated ESN's own $\text{acf}(x^2)$ (blue) sits visibly *above* the real curve (orange) throughout most of the lag range – these 3 stocks' own real squared-return memory decays unusually fast, and the same widened-persistence calibration that correctly matches their $\text{acf}(|x|)$ now sustains more $\text{acf}(x^2)$ memory than they actually show.

4 Interpretation

The persistence ceiling documented in earlier rounds of this calibration was, like the leverage-effect note's own front-loading, a parameter bound rather than a genuine architectural limit – but a different parameter this time: λ_{hi} 's own standard lower edge, 1.0, used without individual scrutiny across every stylised fact calibrated in this note family, was itself preventing the reservoir's fastest mode from decaying slowly enough to sustain the persistence real markets show. Widening it to 0.001 and recalibrating resolves the indices' own persistence gap essentially completely, confirmed directly via a 90% confidence band rather than a single mean curve.

The 3-stock $\text{acf}(x^2)$ overshoot is a genuinely different, and still open, kind of mismatch: it is not that the architecture lacks persistence – if anything, for these 3 assets specifically, it now has more than their own real squared-return series shows. This suggests these 3 stocks’ own return series carry a form of variance clustering unusually short-lived compared to both the rest of this bench and the architecture’s own achievable range at the parameters that otherwise fit them well – a property of these particular assets, not obviously a further failure of the calibration itself, though this was not investigated beyond noting the pattern.

5 Limitations and Recommended Next Steps

1. **Every one of the 9 assets’ own calibrated H lands at or within 0.03 of its own standard upper bound, 0.3,** mirroring exactly the pattern that led to widening λ_{hi} in the first place. Whether H ’s own bound is also too narrow for the Taylor effect specifically was not tested here – a natural, direct next step given the diagnostic already established in §2.3.
2. **The 3-stock $\text{acf}(x^2)$ overshoot (LHX, HUM, KMB) was not resolved.** Freeing a 5th parameter (e.g. m_1 or m_2 , both held fixed throughout, following the leverage-effect note’s own precedent for a level-correcting parameter) might separate the 2 curves’ own persistence independently, rather than the current 4 parameters moving both together, but this was not tested.
3. **λ_{hi} ’s own lower bound, 0.001, was chosen generously rather than tightly:** every asset’s calibrated value falls between 0.037 and 0.064, comfortably above this bound rather than pressed against it, but whether a different asset bench might push closer to it was not examined.
4. **Whether widening λ_{hi} (and possibly H) disturbs any of the other 10 stylised facts calibrated elsewhere in this note family with both parameters held at their original, narrower ranges was not checked** – a full re-run of the companion note’s own 5-asset, 11-stylised-fact calibration with these bounds widened would be needed to know whether this is a strict improvement overall.
5. **The 6 stocks and 3 indices were not chosen for any particular economic or sectoral reason** – the stocks are an independently randomised selection (seed 2026), and the indices are the only 3 with available OHLC data.